Monitor Warning System – General Data

1. General

The Monitor Warning System (MWS) is responsible for all APEX warning and caution functions.

2. Description

A. Monitor Warning System

The Monitor Warning System (MWS) controls the CAS messages, aural warnings, and annunciation. The MWS also records information in the Fault History Data Base (FHDB) via the Central Maintenance Computer (CMC).

The Apex system includes two Monitor and Warning Function (MWF) systems, one in each AIOP module, that independently calculate the logic for Crew Alerting System (CAS) messages. This allows one channel to control the MWS functions, such as aural alerts and CAS messages, while the other channel monitors the controlling channel for redundancy. Comparison monitors alert the pilot when there is difference between the results generated from MWF system 1 and MWF system 2.

MWF system 1 displays DAU channel A CAS messages and MWF system 2 displays DAU channel B CAS messages. Both channels of a DAU should always output the same message but in the event of a malfunction the two DAU channels could disagree and trigger the miscompare monitor.

In normal operation there should be no discrepancy between the calculations of the two MWF systems. In the event of a fault in a DAU channel that results in different outputs between DAU channels, the CAS miscompare monitor will activate. The pilot may then toggle the CAS message window between MWF system 1 and MWF system 2 by pressing the bezel button beside the small amber ‘MW’ square to display and compare output from each of the two MWF systems. By identifying the difference between the CAS messages generated by the two MWF system, and then gathering further information about the affected aircraft component from other sources (e.g. engine indications, mechanical trim indications, visual observation of flap or control surface positions, etc.), it should be possible to determine which MWF system is most credible. The selected channel is annunciated by the larger number in the 1/2 indication directly below the MW square.

Under certain rare circumstances (for example, an advisory maintenance message concerning one of the two DAUs) both MWF systems will be correct. The difference between the MWF systems will trigger a MW annunciation because the DAU maintenance message sent to each individual MWF will contain fractionally different text (DAU channel A self-identification sent to MWF 1 will differ from DAU channel B self-identification sent to MWF 2).

B. Crew Alerting System (CAS)

The Crew Alerting System (CAS) provides visual warning to the crew regarding malfunctions or abnormal conditions which occur during operation.

The CAS messages are grouped by severity into four priority levels: warning messages, which are displayed in red text, caution messages, which are displayed in amber text, advisory messages, which are displayed in cyan text, and status messages, which are displayed in white text.

CAS messages are posted in the lower left 1/6 window of the system window on the multi-function display.

Warning messages are accompanied by illumination of the red MASTER WARNING lights and sounding of a triple chime. Caution messages are accompanied by illumination of the amber MASTER CAUTION lights and sounding of a single chime. Advisory and status messages are not annunciated.

Refer to the Section 7 of the Pilot Operating Handbook, PSM 1-64-POH for additional details on the CAS messages.

Some parameters are annunciated on the ADI instead of with a CAS message. These include STALL, FIRE, NO TAKEOFF, and SPEED warnings.

C. Aural Warning System

Various aural warnings are generated by the Apex avionics system. Each aural alert is prioritized and grouped with other audio alerts, called priority groups, as is shown in Section 7 of the Pilot Operating Handbook, PSM 1-64-POH. Group 1 is the highest priority group and Group 4 is the lowest priority group. When two alerts of different groups are enabled at the same time, only the aural alerts of the highest priority group will be sounded. When the alerts of the highest priority group stop, the aural alert of the next highest priority group will be sounded, if still active.

Aural warnings listed as 'mutable' in the AFM are silenced by pressing the Master Caution (MC) or Master Warning (MW) switch, as appropriate to the notification. When the logic that drives a mutable aural is enabled and the appropriate MC/MW switch is depressed, the mutable aural is silenced until the enabling logic transitions off. When multiple mutable aural warnings are active, all will be muted at once. Therefore, each aural cannot be muted individually. All aural warnings will be heard at least once before muting becomes effective.

Aural warnings are independently processed by both channels of the custom I/O board in the MAU, then compared to ensure validity. In the event of a passive failure of one channel, an Aural Warning Fault advisory (cyan) CAS message advising of the failure will be displayed, but full aural warning capability will continue to be available from the single remaining channel. In the event of an active failure of one channel that results in unwanted sound coming from one of the two channels, the pilot can silence the defective channel by selecting the AURAL 1 MUTE or AURAL 2 MUTE switch, as appropriate, to the MUTE position. If both switches are placed in the MUTE position, no aural warnings will be heard and an Aural Warning Fail caution (amber) CAS message will be displayed.

TAWS and TCAS aural warnings are separate from the Apex aural warnings and are connected to the pilot's and co-pilot's audio panels through an audio mixer, refer to 23-50-00, Audio Integration – General Data for details.

D. Master Warning and Caution Lights

Two 1 inch by 1.2 inch LED switchlights, internally labelled MASTER CAUTION and MASTER WARNING, are provided directly above the pilot's and co-pilot's Primary Flight Displays.

These lights will illuminate whenever a caution level (amber) CAS message or a warning level (red) CAS message appears. Pressing the light provides a ground to the MAU which acknowledges the alert. This will extinguish the light and remove the highlight from all new CAS messages. The caution lights are powered by the Caution Lights Dimming Control box, and triggered by a ground from the MAU. Refer to 33-10-11, Caution Lights Dimming Control Box – General Data for more information.

3. Adjustment/Test

A. Monitor Warning System (MWS)

The MWS has built in test and fault detection and exclusion capability. Failures of components or loss of system capability will be announced to the crew using the aircraft crew alerting system. Information may be presented as caution or advisory messages, depending on the importance of the message or the severity of the condition.

B. Master Warning and Caution Lights Function Test

1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) Open HYDRAULIC OIL PUMP (C6) circuit breaker to trigger the Master Warning annunciation and a highlighted Hydraulic Power Fail warning CAS.

3) Press the pilot's master warning pushbutton and confirm the warning annunciation clears and CAS is no longer highlighted.

4) Open FUEL CROSSFEED (D7) circuit breaker to trigger the Master Caution annunciation and a highlighted Fuel Crossfeed Fail caution CAS.

5) Press the pilot MASTER CAUTION pushbutton and confirm the Caution annunciation clears and CAS is no longer highlighted.

6) Repeat Para 3.B.2)to Para 3.B.5) for co-pilot's side light.

7) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

4. Removal and Installation

A. Removal of Master Warning and Caution Lights

1) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

2) Open CAUTION LTS NORMAL (N3) and CAUTION LTS BYPASS (N2) circuit breakers on the avionics circuit breaker panel and fit safety clips.

3) Gain access to the rear of the lights by removing the pilot's or co-pilot's PFD, as required. Refer to 31-11-00, Display Units – General Data.

4) Disconnect electrical connector.

5) Using the extraction slots, pull the pushbutton cap out of the switch body until it rotates down.

6) Loosen the (2) internal screws until the locking pawls release and rotate into the switch body.

7) While holding the mounting sleeve behind the panel, pull the switch body out of the panel from the front.

8) If aircraft power is required with the unit removed, ensure the connector is stowed as required.

B. Installation of Master Warning and Caution Lights

1) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

2) Ensure the CAUTION LTS NORMAL (N3) and CAUTION LTS BYPASS (N2) circuit breakers on the avionics circuit breaker panel are open and fitted with safety clips.

3) Using the extraction slots, pull the pushbutton cap out of the switch body until it rotates down.

4) Ensure the locking pawls are fully retracted by loosening the (2) internal screws.

5) Insert the switch body though the panel from the front.

6) Slide the mounting sleeve over the switch body from the rear of the panel.

7) Tighten (2) internal screws until the locking pawls pull the mounting sleeve tight against the panel.

8) Insert the pushbutton cap into switch body.

9) Connect the electrical connector.

10) Install pilot's or co-pilot's PFD, as required. Refer to 31-13-00, Apex Controllers – General data.

11) Remove safety clips and close CAUTION LTS NORMAL (N3) and CAUTION LTS BYPASS (N2) circuit breakers on the avionics circuit breaker panel.

12) Do a function test of the lights, refer to Para 3.B.

5. Troubleshooting

A. Monitor Warning System (MWS)

The monitor warning system is contained within the MAU AIOP module, any errors will be annunciated with a CAS message. Refer to 31-43-00, Modular Avionics Unit (MAU) (Mod 6/2049) – General Data for information on the AIOP module.

If a system CAS message is present, refer to the applicable systems Chapter from this Maintenance Manual for troubleshooting information.

B. Master Warning and Caution Lights

1) Ensure CAUTION LTS NORMAL (N3) and CAUTION LTS BYPASS (N2) circuit breakers are closed.

2) Remove electrical connectors and visually inspect the connectors for damaged wiring or contacts.

3) Ensure connectors are fully seated and tight.

4) If troubleshooting at a wire level is required, refer to the applicable wiring diagrams in Wiring Manual PSM 1-64-2W.

5) Swap pilot and co-pilot lights.